

Supplementary Material for WorldRoll: From Privileged Camera Trajectories to Pose-Free World Rollouts

Anonymous submission

Related Work

World models and image-to-video priors. Classical world models learn predictive latent dynamics for planning (Ha and Schmidhuber 2018; Hafner et al. 2020); recent generative systems instead synthesize visually rich futures conditioned on actions, robot trajectories, vehicle controls, or camera commands. Genie, GameNGen, GAIA-1, IRASim, Cosmos, and HY-WorldPlay illustrate this progression from learned environments to interactive video simulation (Bruce et al. 2024; Valevski et al. 2025; Hu et al. 2023; Zhu et al. 2025; NVIDIA et al. 2025; Sun et al. 2025). In parallel, Stable Video Diffusion, I2VGen-XL, VideoCrafter, DynamicCrafter, LTX-Video, Wan, and CogVideoX have advanced appearance and motion priors for video generation, including model families that support or can be adapted to first-frame-conditioned generation (Blattmann et al. 2023; Zhang et al. 2023; Chen et al. 2023, 2024a; Xing et al. 2024; HaCohen et al. 2024; Wan Team 2025; Yang et al. 2025). WorldRoll does not introduce another backbone; instead, it transfers control into two existing compact I2V families through low-rank updates (Hu et al. 2022) while retaining each model’s native diffusion or flow-matching target (Lipman et al. 2023).

Camera control, view synthesis, and data-centric transfer. MotionCtrl, CameraCtrl, and CamI2V condition video diffusion on explicit object- or camera-motion signals, including camera trajectories (Wang et al. 2024; He et al. 2025; Zheng et al. 2024). Geometry-assisted methods additionally render or reconstruct a scene: ViewCrafter conditions on point-based renders, RealCam-I2V exposes an interactive 3D trajectory, and TrajectoryCrafter and ReCamMaster redirect monocular video (Yu et al. 2025b; Li et al. 2025b; Yu et al. 2025a; Bai et al. 2025). These methods pursue accurate execution of an explicit test-time path. WorldRoll uses related geometric signals only to construct supervision, preserving the student’s ordinary image-and-text interface at test time. This transfer depends on data quality. OpenVid-1M and Panda-70M demonstrate the value of filtering and multimodal recaptioning (Nan et al. 2025; Chen et al. 2024b), RealEstate10K provides aligned camera tracks (Zhou et al.

2018), and DL3DV-10K supplies diverse real viewpoints for teacher initialization (Ling et al. 2024). Accordingly, the TG stream performs offline dataset distillation: teacher videos become ordinary training examples, without logit, feature, score, or diffusion-trajectory matching.

Evaluation. WorldScore measures static and dynamic properties of generated worlds, and WorldModelBench emphasizes instruction following, physical adherence, and common sense (Duan et al. 2025; Li et al. 2025a). These two benchmarks form the main evaluation because they are designed to judge generated videos as worlds.

Construction and Training Details

Data Construction

Normalization and camera language. Every experimental clip spans five seconds, and curation uses 121 frames at 24 fps. Real clips are center-cropped to 512^2 ; HY-WorldPlay outputs remain at 640^2 in the accepted pool. Student caches then resample media to each backbone’s native grid. Figure 1 summarizes the 19-action camera ontology in the five categories used throughout the paper: stationary, optical, translation, rotation, and curved. A segmented camera path contains one to four contiguous segments that cover the clip: actions within a segment are simultaneous, whereas successive segments are sequential. The validator rejects paths with gaps, overlaps, zero-length segments, or `static` combined with motion. Pose-supported rotations below 5° are treated as drift; larger rotations are rounded to readable values, and translations use one significant digit in clip-local units. Magnitudes are included only when visible motion and camera-trajectory statistics agree.

Real-video prefiltering. Source metadata is used only to remove clearly unsuitable DR candidates before the common VLM annotation and filtering stages. OpenVid candidates must be at least five seconds long and meet the released aesthetic and temporal-consistency thresholds of 5.5 and 0.95, respectively. Panda candidates must meet a released video-text matching threshold of 0.43, be marked desirable, and contain no detected multi-shot edit. Released captions and camera tags are not used as WorldRoll annotations; the selected clip remains the authority for scene content, independent subject motion, and camera motion.

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Pose evidence and VLM annotation. For RealEstate10K, each world-to-camera matrix is inverted to obtain T_t ; for a candidate span $[a, b]$, the relative transform is $\Delta T_{a \rightarrow b} = T_a^{-1} T_b$. We summarize camera-center distance traveled, start-to-end displacement, their ratio, cumulative and endpoint orientation change, and signed translation and yaw/pitch/roll in the starting camera’s right-, down-, and forward-positive axes. Together, the distance ratio and net-versus-cumulative rotations expose reversals or curved motion hidden by endpoint transforms. For PA, Qwen3.6-27B (Qwen Team 2026b) receives 20 frames sampled at 4 fps and the camera-trajectory statistics; for DR, it receives only the video. Both settings use deterministic, schema-constrained decoding. The VLM returns a persistent scene description, optional independent subject motion, and a segmented camera path; deterministic code then renders the canonical prompt. Whole-clip statistics are primary quantitative evidence. Coarse interval statistics may support a visually chosen segment but never determine a boundary, and numeric evidence alone cannot establish zoom, arc, or orbit.

Teacher paths. For teacher generation, HunyuanVideo-1.5 with HY-WorldPlay (Tencent Hunyuan Foundation Model Team 2025; Sun et al. 2025) receives a real source frame, scene conditioning, and a sampled camera path converted to its native control representation; the text describes the scene but omits the path. The library contains nine single-segment commands—*static*, *push/pull*, *truck left/right*, *pan left/right*, and *tilt up/down*—and eight translation-then-pan paths. Single- and two-segment families are sampled with probabilities 0.65 and 0.35, respectively, followed by uniform sampling within the selected family. For a two-segment path, the verbalized segment boundaries over the five-second clip follow the teacher’s control allocation. Teacher paths remain categorical: the request is exact, but realized motion strength is verified visually rather than assigned an estimated magnitude. After generation, the same path is rendered into the student-facing canonical prompt.

VLM quality gate. The quality pass is separate from annotation. It receives the full clip, canonical prompt, segmented camera path, and camera-trajectory statistics when available, and returns one Boolean for each applicable fatal or quality check. The VLM does not output an acceptance decision or numeric score. Table 1 gives the complete inventory. Any false fatal check rejects the candidate; if all fatal checks pass, the Boolean quality criteria are averaged within groups and then across groups, and the candidate is retained when the resulting score is at least 0.8. Quality affects admission and subset ranking but never the training loss.

Curation Prompts and Worked Examples

Table 5 reproduces the VLM templates used at the two model-judged stages. PA and DR share the annotation template, with the trajectory-evidence addendum used only for PA; TG instead sends its known path and grounded scene description directly to the deterministic composer. All streams use the filtering template with only applicable checks instantiated. The runtime enforces schema-constrained JSON

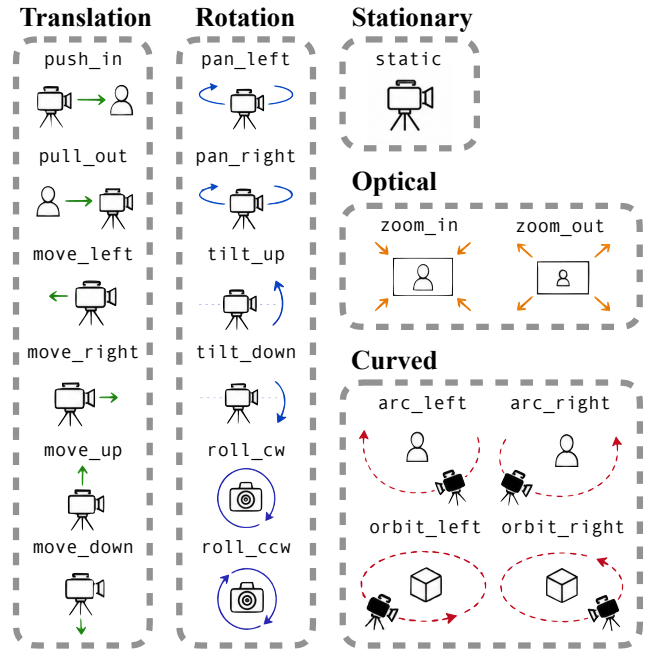


Figure 1: **Camera-action ontology.** The 19 actions are grouped into five categories: *stationary* contains *static*; *optical* contains *zoom_in* and *zoom_out*; *translation* contains *push/pull*, *lateral truck*, and *vertical move* actions; *rotation* contains *pan*, *tilt*, and *roll* actions; and *curved* contains *arc* and *object-centric orbit* actions. *Push/pull* denotes physical translation rather than zoom, *arcs* require curved translation with yaw, and *orbits* require visible motion around a subject or coherent scene region. Actions shown in one temporal segment are simultaneous; successive segments are sequential.

output and stores no free-form chain-of-thought.

Pose-grounded example. A five-second kitchen clip shows a coherent rightward turn. The whole-clip evidence reports 58.1° rightward yaw, 0.7° downward pitch, -0.3° clockwise roll, and 0.083 clip-local units of travel. The VLM emits a bright-kitchen description, a null subject-motion field, and a full-interval *pan_right* with a rounded 60° magnitude, without promoting the subthreshold pitch and roll or the small translation. The deterministic composer produces:

Start from the input image. Scene: A bright kitchen has white cabinets, a central wooden island, and pendant lights. The camera pans right 60 degrees. Continue as one coherent shot with stable geometry, appearance, identities, and lighting. Reveal only adjacent parts of the same environment; no cuts, jumps, or unrelated elements.

Filtering example. Consider a fixed-view park clip whose canonical prompt describes a brown dog running left to right. The clip is semantically correct but exhibits visible grass-compression artifacts and a brief brightness flicker. The filter returns *true* for each of the five fatal checks, *false* for the compression and flicker quality checks, and *true* for every

Group	Applies to	Checks returned by the VLM
<i>Fatal checks</i>		
Core validity	PA, DR, TG	Video and first frame are usable; the clip is one continuous shot; the clip remains one coherent world; the canonical prompt is fundamentally grounded; and the dominant camera action, direction, and temporal order are fundamentally correct. Uncertainty is a failure.
Pose compatibility	PA	Visible global motion agrees with the aligned camera-trajectory statistics; every attached magnitude is broadly compatible with both the clip and the statistics.
<i>Quality checks</i>		
Visual fidelity	PA, DR, TG	First-frame clarity, sharpness, exposure, stable natural color, acceptable compression and overlays, acceptable distortion, and sufficient visible detail.
Temporal fidelity	PA, DR, TG	Smooth motion and pacing; no material freeze, stutter, flicker, jitter, ghosting, or temporal warping; stable focus and exposure.
World consistency	PA, DR, TG	Stable geometry, object appearance, object permanence, and coherent lighting; when relevant, adjacent revealed regions and plausible parallax/depth ordering.
Scene grounding	PA, DR, TG	Named content, appearance, and layout are visible and persistent; no unsupported objects; description matches both first frame and continuation.
Camera control	PA, DR, TG	Dominant action and segment timing; when relevant, direction, translation-versus-rotation, optical-versus-physical motion, simultaneous execution, sequential order, static intervals, and arc/orbit geometry.
Subject-motion grounding	Annotated motion	The subject is visible; every described motion and its timing match camera-independent motion in the clip.
Dynamic-motion quality	DR	Independent motion is clear, meaningful, coherent, identifiable, sufficiently visible, and separable from camera motion.
Generation fidelity	TG	Requested path execution, start-scene preservation, stable foreground/background structure, limited generation artifacts, and temporally consistent camera response.
Trajectory alignment	PA	Global motion and semantic path agree with the statistics; sub-threshold rotation is not promoted; applicable translation and rotation directions match.
Magnitude accuracy	PA with magnitudes	Every magnitude is plausible and agrees with the statistics over its own annotated segment.

Table 1: **Structured VLM filtering checks.** Fatal checks enforce minimum validity and reject a candidate immediately when any Boolean is false. Quality checks are instantiated only when applicable, averaged within each semantic group, and then averaged across groups so that long checklists cannot dominate smaller control-critical groups. A candidate is accepted only when all fatal checks pass and the group-balanced quality score is at least 0.8.

other instantiated check. The seven group scores are 0.875, 0.875, 1, 1, 1, 1, and 1, giving $Q = 0.964$; deterministic code therefore accepts the clip. A false single-shot or fundamental camera-path check would cause rejection regardless of Q . This example illustrates why the VLM returns criterion-level Booleans and reasons instead of making an acceptance decision.

Student Adaptation and Inference

Table 2 gives the two native contracts and the shared main-run optimization settings. All runs use LoRA, bfloat16 precision, AdamW, a constant learning rate of 10^{-5} , and an effective batch size of 16. LTX-Video excludes clean first-frame latent tokens from the flow loss. CogVideoX-Fun uses a first-frame-only inpainting mask and its native masked diffusion objective (aigc-apps 2026). No student-ID token, pose encoder, action adapter, or auxiliary camera loss is used.

Qwen3.5-2B (Qwen Team 2026a) greedily rewrites free-form instructions without access to the image; when scene content is absent, it inserts “the scene shown in the input image.” Base $_{\mathcal{R}}$ and WorldRoll use the same cached rewrite. During multi-chunk rollout, chunk k uses seed $s + k$, and its

final frame initializes chunk $k + 1$. The duplicated boundary frame is removed during concatenation, and instructions may change between chunks.

Evaluation and Dataset Diagnostics

Evaluation protocol and baselines. WorldScore reports static- and dynamic-world aggregates and ten components; WorldModelBench reports instruction following on a 0–3 scale, physical adherence on a 0–5 scale, common sense on a 0–2 scale, and their 0–10 sum. All three local variants of a student share benchmark cases and seeds, and the two rewritten variants share one prompt cache. Public rows provide benchmark-specific context rather than matched reproductions because their interfaces and sampling budgets differ. The reported public results use dated official leaderboard snapshots and list exact model revisions wherever available. The data-composition experiment uses WorldModelBench and the two starred WorldScore subset aggregates, whereas the adapter-capacity, data-scale, and training-duration experiments use WorldModelBench alone.

Specification	LTX-Video	CogVideoX-Fun
<i>Model-specific</i>		
Parameter scale	2B	2B
Checkpoint	v0.9.6 dev	V1.1 2B InP
Frames / fps	121 / 24	41 / 8
First frame condition	Clean latents	Inpainting mask
Prediction target	Flow matching	Masked diffusion
Loss on first frame	Tokens excluded	Native inpainting
<i>Shared</i>		
Clip duration	5 s	
Training resolution	512 × 512	
Student text input	Canonical prompt only	
Numeric precision	bfloat16	
Adapter	LoRA	
LoRA rank / α	128 / 128	
Optimizer	AdamW	
Effective batch size	16	
Learning rate	Constant 10^{-5}	
Optimizer steps	20K	

Table 2: **Student-model specifications.** The upper block contrasts the native contracts of the two approximately 2B-parameter students; the lower block consolidates their shared main-run adaptation settings. Both models receive only the first frame and canonical prompt, generate five-second clips on their native temporal grids, and retain their native conditioning and prediction targets during LoRA fine-tuning.

Data (K)			Selected samples				
PA	DR	TG	PA-RE	DR-OV	DR-P70	TG-DL	TG-RE
0	0	10	0	0	0	5,000	5,000
5	0	5	5,000	0	0	4,000	1,000
5	5	0	5,000	4,000	1,000	0	0
0	5	5	0	4,000	1,000	2,500	2,500
2	3	5	2,000	2,400	600	4,000	1,000
3	3	4	3,000	2,400	600	3,200	800
6	6	8	6,000	4,800	1,200	7,000	1,000
12	12	16	12,000	9,600	2,400	12,000	4,000
20	20	20	20,000	16,000	4,000	14,000	6,000

Table 3: **Exact training views.** The left block gives the PA/DR/TG composition in thousands; the right block gives the exact number of selected samples from the five stream-specific sources after filtering. RE, OV, P70, and DL abbreviate RealEstate10K, OpenVid-1M, Panda-70M, and DL3DV-10K, respectively. The 60K 20+20+20 view is used for the main results.

Dataset Composition and Diagnostics

Exact training views. Table 3 records the source-level composition of every data view reported in the paper. PA samples always come from RealEstate10K. DR preserves a 4:1 OpenVid-1M-to-Panda-70M ratio at every scale, whereas the TG source ratio varies as shown.

Camera-path statistics. Table 4 characterizes path structure, temporal composition, label diversity, and magnitude annotations of the three streams in the main training view.

Statistic	Unit	PA	DR	TG
<i>Path-level</i>				
Segments per path	mean	1.00	1.04	1.37
Actions per path	mean	1.45	1.05	1.37
Segment duration	median s	5.00	5.00	2.58
Multi-segment paths	%	0.4	3.5	36.8
Compound paths	%	43.7	0.6	0.0
Static-only paths	%	0.1	90.7	7.9
Magnitude paths	%	98.7	—	—
Observed transitions	/361	30	70	8
Transition entropy	[0, 1]	0.496	0.529	0.353
<i>Action-level</i>				
Observed labels	/19	18	17	9
Label entropy	[0, 1]	0.546	0.184	0.700
Effective label count	[1, 19]	4.99	1.72	7.86
Stationary share	%	0.1	89.8	5.8
Optical share	%	0.0	0.6	0.0
Translation share	%	47.0	3.5	48.5
Rotation share	%	52.1	6.0	45.7
Curved share	%	0.9	0.1	0.0
Rotation coverage	%	99.8	—	—
Translation coverage	%	99.8	—	—
Rotation magnitude	◦	30 [15–45]	—	—
Translation magnitude	local	3 [2–3]	—	—

Table 4: **Camera-path composition by training stream.** Statistics are computed over the exact 60K main training view. An action instance is one label in one segment; a multi-segment path contains sequential motion, and a compound path contains at least one segment with simultaneous actions. Transitions are ordered label pairs across adjacent segments; their coverage and entropy are computed against the $19^2 = 361$ possible pairs, with entropy normalized by $\log 361$. Label entropy is normalized by $\log 19$, and the effective label count is the exponential of the corresponding unnormalized entropy. Family shares use all action instances as their denominator. Magnitude coverage is measured among eligible PA rotation or translation instances; magnitude summaries report median [IQR] while keeping degrees and clip-local translation units separate. Dashes denote annotations unavailable by construction.

Stream complementarity. The three streams exhibit distinct and complementary path structures. PA is predominantly single-segment but frequently simultaneous: only 0.4% of paths contain multiple segments, whereas 43.7% contain compound motion and the mean path contains 1.45 actions. Its action instances divide almost evenly between translation (47.0%) and rotation (52.1%), and 98.7% of paths carry at least one grounded magnitude. Magnitude coverage reaches 99.8% for both eligible classes, with a median rotation of 30° [15–45] and a median clip-local translation of 3 [2–3]. DR provides the opposite regime: 90.7% of paths are static-only and stationary instances account for 89.8% of its actions, consistent with this stream emphasizing independent scene and subject dynamics rather than viewpoint motion. Although DR observes 17 labels, its effective count

is only 1.72 because this stationary mode dominates. TG supplies the strongest sequential structure: 36.8% of paths are multi-segment, the pooled median segment duration is 2.58 seconds, and no paths are compound. Its nine-label teacher library yields the highest normalized label entropy (0.700) and an effective count of 7.86. Its eight observed transition types are also nearly uniformly represented, as indicated by a transition entropy of 0.353, the maximum normalized value for eight equally frequent pairs. Thus, PA contributes simultaneous geometry-grounded control, DR contributes mostly fixed-view dynamics, and TG contributes deliberately balanced sequential camera paths.

Reproducibility and release. The reproducibility artifact records model revisions, prompt caches, threshold, source splits, subset manifests, LoRA and optimizer settings, hardware, gradient accumulation, wall-clock time, seeds, and per-case outputs. Release scope follows upstream licenses: construction and evaluation code, prompts, provenance/quality metadata, subset manifests, and LoRA weights can be provided; media are redistributed only where permitted.

Discussion

Control–fidelity trade-off. The two benchmarks expose complementary parts of the same transfer. Canonical rewriting alone helps both students on WorldModelBench but has a backbone-dependent effect on WorldScore, indicating that clearer language can reduce semantic ambiguity without guaranteeing that a pretrained video model will execute the requested camera path. Fine-tuning closes this execution gap for both architectures: the shared gains in camera control, content alignment, motion accuracy, and both benchmark aggregates show that WorldRoll transfers more than a preferred prompt style. At the same time, reductions in photometric consistency, motion smoothness, and motion-magnitude accuracy reveal a genuine control–fidelity frontier. The present objective rewards all accepted videos uniformly, so examples with salient camera motion can pull the students away from conservative continuations favored by their pretrained priors. Mixing a small replay set from the original pretraining distribution, reweighting static and low-motion examples, or selecting checkpoints with a multi-metric criterion may improve this frontier.

Data complementarity and scaling. The data sources explain why no single composition dominates. PA supplies the strongest evidence for direction and magnitude, but RealEstate10K is comparatively narrow in scene type and often contains limited independent subject motion. DR broadens appearance and natural dynamics through OpenVid-1M and Panda-70M, although camera annotations inferred from pixels are necessarily noisier. TG deliberately covers targeted single- and multi-segment paths, but it is bounded by the teacher’s action library and can reproduce teacher-specific artifacts. The 3:3:4 mixture is therefore best understood as a robust compromise for the controlled 10K studies rather than a universal optimum: it retains pose-grounded camera semantics while allocating most examples to broader real dynamics and targeted path coverage. The main 60K view instead scales all three streams to 20K

examples each. The rank and scale ablations support the same data-centric interpretation—moderate adapter capacity is sufficient across backbones, whereas additional unique curated examples continue to help—but the late-step regression shows that data scale alone does not remove the need for learning-rate scheduling, larger effective batches, or validation-based early stopping.

Real-Video Annotation (PA and DR)

INPUT AND TASK

[VIDEO: one uniformly sampled, continuous five-second clip]

Annotate this video for first-frame-conditioned world rollout. The first clip frame is the rollout input image.

Use the frames to determine: (1) persistent scene content, (2) motion independent of the camera, and (3) the camera path. Return only claims supported by the video.

CAMERA VOCABULARY

{CAMERA_ONTOLOGY}

RULES

- Write one or two concrete scene sentences.
- Describe only camera-independent subject motion; otherwise return null.
- Divide 0.0-5.0 seconds into 1-4 contiguous camera segments.
- Actions within a segment are simultaneous; successive segments are sequential.
- Use the fewest visually supported segments; merge shake, drift, and brief deviations.
- Infer camera motion from global flow, parallax, occlusion, and revealed regions, not local object flow.
- Separate subject and camera motion.
- Distinguish push/pull from optical zoom; use arc only for curved translation with yaw and orbit only for motion around a visible subject or region.
- static must occur alone.

OUTPUT CONTRACT

Return JSON only:

```
{"scene":...,
"subject_motion":null or ...,
"camera_path":[
{"start":0.0,"end":5.0,
"actions":[{"action":...}]}]}
```

Quality Filtering (PA, DR, and TG)

INPUT AND TASK

[VIDEO: one uniformly sampled, continuous five-second clip]

Judge this video for first-frame-conditioned world-rollout training. The first clip frame is the rollout input image.

RULES

- A fatal check is true only when the clip clearly passes it; uncertainty means false.
- A quality check is true only when the candidate is solidly usable on that criterion.
- Return every listed check exactly once; do not add or omit checks.
- Use the frames as authority for visible content and motion.
- Camera motion is global viewpoint change; subject motion is camera-independent motion.
- Push/pull is not zoom; arc requires curved translation plus yaw; orbit requires motion around a subject or region.
- Do not output acceptance or a numeric score.

CANDIDATE

```
{"prompt":{CANONICAL_PROMPT},
"camera_path":{STRUCTURED_PATH},
"annotation":{"scene":{SCENE},
"subject_motion":{MOTION_OR_NULL}}}
{OPTIONAL_PA_TRAJECTORY_EVIDENCE}
```

REQUIRED CHECKS

Fatal: {FATAL_CHECK_DICTIONARY}
Grouped quality:
{APPLICABLE_GROUPED_CHECK_DICTIONARY}

OUTPUT CONTRACT

Return JSON only; replace each null with true or false:

```
{"fatal_checks":{...},
"quality_checks":{
"group":{"criterion":null}},
"reasons":[]}
```

PA-ONLY EVIDENCE ADDENDUM

Keep the video primary for action meaning and segment boundaries. Treat whole-clip measurements as primary and interval measurements as secondary. Translations use clip-local units and rotations use degrees. Net axis change below 5° is drift; accumulated jitter is not an action. Measurements alone cannot establish zoom, arc, or orbit. Emit a positive, rounded magnitude only when video and measurements agree; otherwise emit null.

{TRAJECTORY_EVIDENCE_JSON}

Table 5: **Curation and filtering prompt templates.** Braced fields are expanded programmatically using the ontology in Figure 1 and the applicable checks in Table 1. Gray labels are presentational; the VLM returns structured observations or criterion-level Booleans, never a final prompt, decision, or score.

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